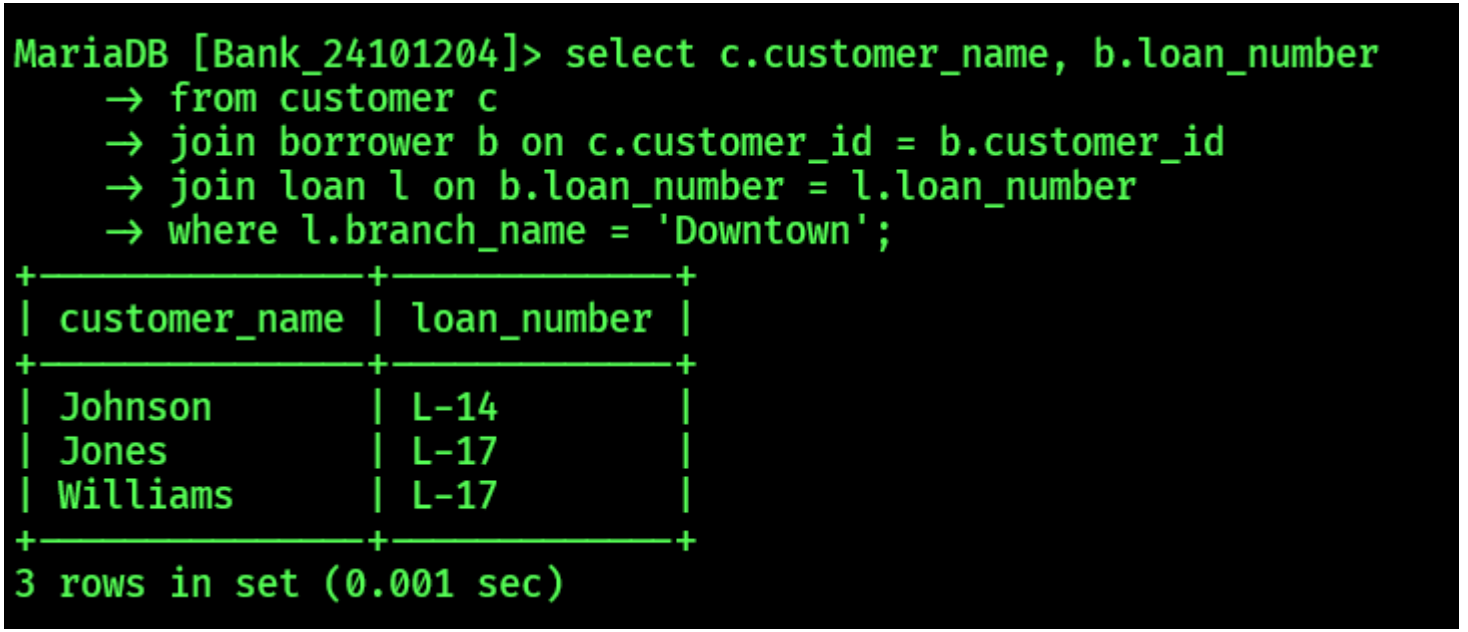
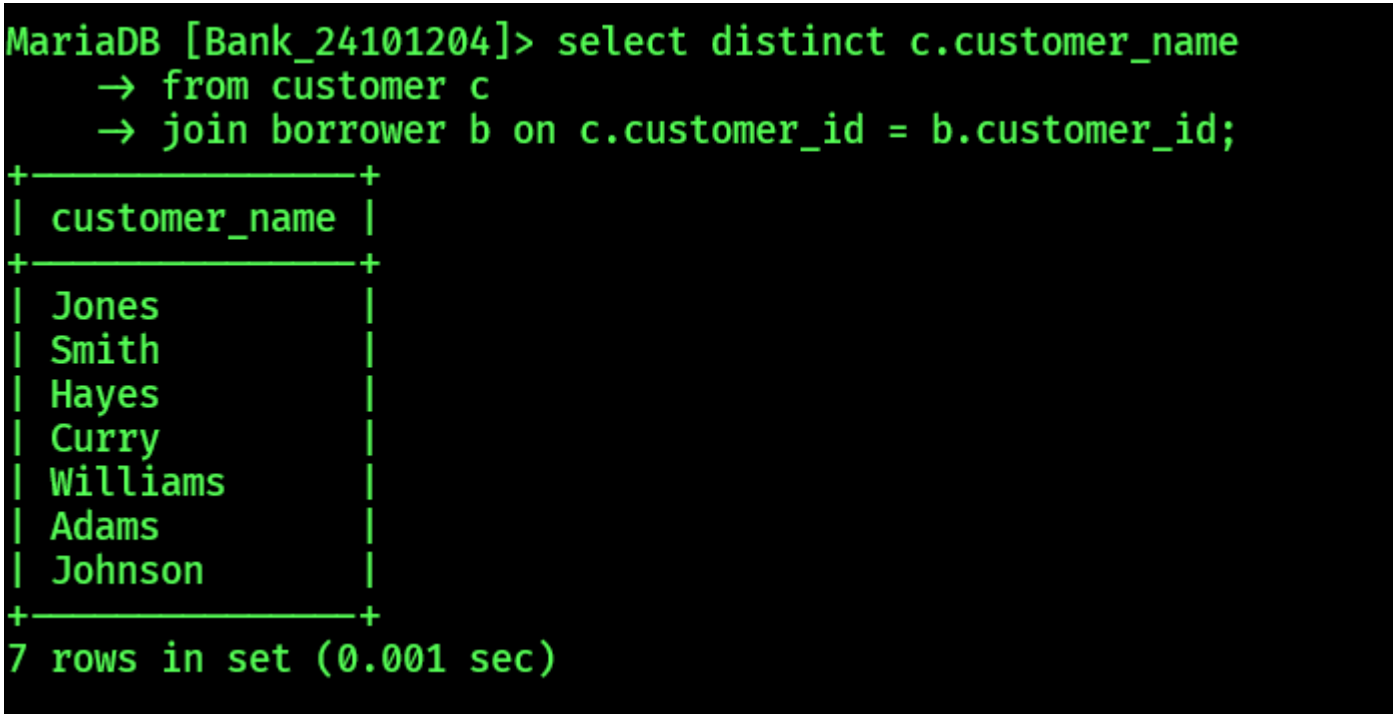
CSE370: Database Systems

Assignment 03 | Spring 2026

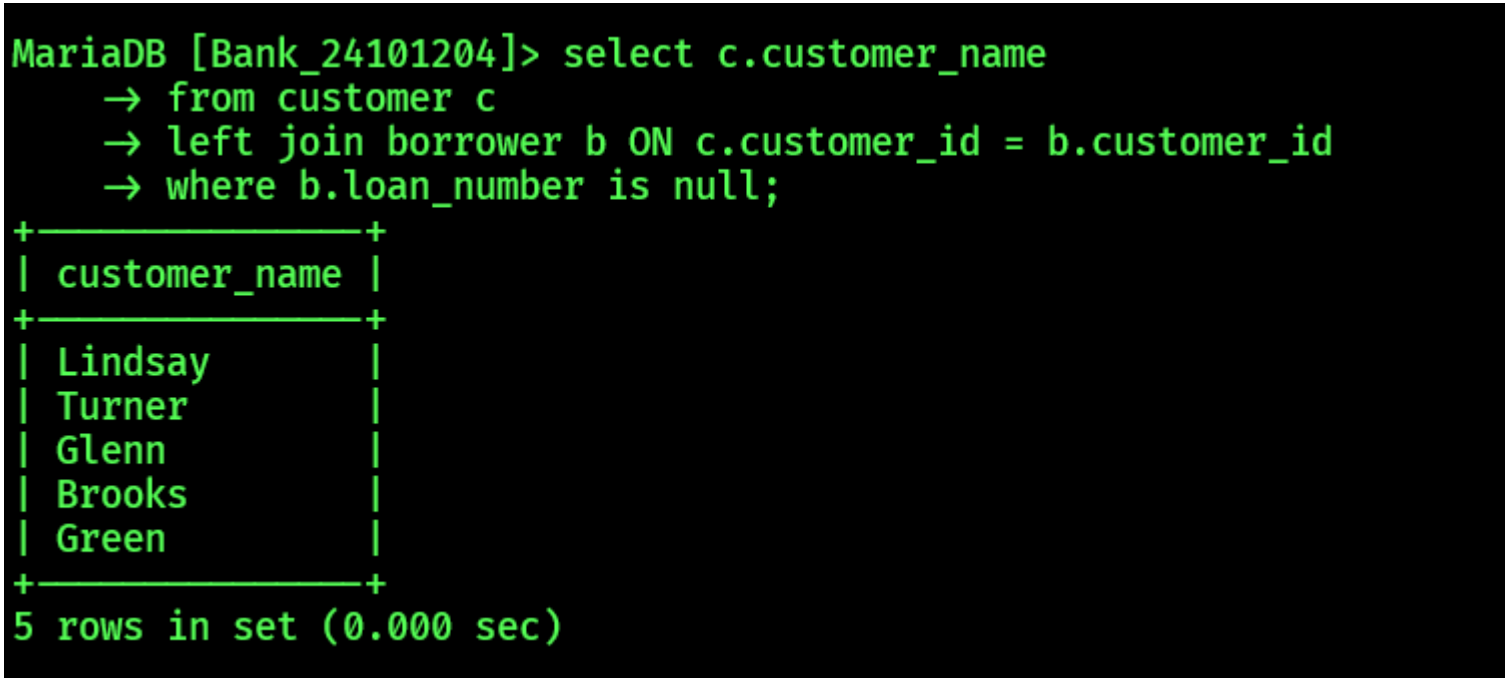
No 1 Query (as Plain Text)	`select c.customer_name, b.loan_number`
	`from customer c`
	`join borrower b on c.customer_id = b.customer_id`
	`join loan l on b.loan_number = l.loan_number`
	`where l.branch_name = 'Downtown';`

No 1 SS (of Query & Output in Shell)	

No 2 Query (as Plain Text)	`select distinct c.customer_name`
	`from customer c`
	`join borrower b on c.customer_id = b.customer_id;`

No 2 SS (of Query & Output in Shell)	

No 3 Query (as Plain Text)	`select c.customer_name`
	`from customer c`
	`left join borrower b ON c.customer_id = b.customer_id`
	`where b.loan_number is null;`

No 3 SS (of Query & Output in Shell)	

No 4 Query (as Plain Text)	`select c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city as City`
	`from customer c1`
	`join customer c2`
	`on c1.customer_city = c2.customer_city`
	`and c1.customer_id < c2.customer_id;`

No 4 SS
(of Query & Output in Shell)

```
MariaDB [Bank_24101204]> select c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city as City
    -> from customer c1
    -> join customer c2
    -> on c1.customer_city = c2.customer_city
    -> and c1.customer_id < c2.customer_id;
+-----------+-----------+------------+
| Customer1 | Customer2 | City       |
+-----------+-----------+------------+
| Jones     | Hayes     | Harrison   |
| Smith     | Curry     | Rye        |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford   |
+-----------+-----------+------------+
4 rows in set (0.001 sec)
```

No 5 Query
(as Plain Text)

```
SELECT a.branch_name,
Round(Sum(a.balance * 0.04), 2) AS Total_Interest
from account a
group by a.branch_name;
```

No 5 SS
(of Query & Output in Shell)

```
MariaDB [Bank_24101204]> SELECT a.branch_name,
    -> Round(Sum(a.balance * 0.04), 2) AS Total_Interest
    -> from account a
    -> group by a.branch_name;
+-------------+----------------+
| branch_name | Total_Interest |
+-------------+----------------+
| Brighton    |          66.00 |
| Downtown    |          20.00 |
| Mianus      |          28.00 |
| Perryridge  |          16.00 |
| Redwood     |          28.00 |
| Round Hill  |          14.00 |
+-------------+----------------+
6 rows in set (0.001 sec)
```

No 6 Query
(as Plain Text)

```
select br.branch_city, a.account_number, a.balance
from account a
join branch br on a.branch_name = br.branch_name
where a.balance = (
    select Max(a2.balance)
    from account a2
    join branch br2 ON a2.branch_name = br2.branch_name
    where br2.branch_city = br.branch_city
);
```

No 6 SS
(of Query & Output in Shell)

```
MariaDB [Bank_24101204]> select br.branch_city, a.account_number, a.balance
    -> from account a
    -> join branch br on a.branch_name = br.branch_name
    -> where a.balance = (
    ->     select Max(a2.balance)
    ->     from account a2
    ->     join branch br2 ON a2.branch_name = br2.branch_name
    ->     where br2.branch_city = br.branch_city
    -> );
+-------------+----------------+---------+
| branch_city | account_number | balance |
+-------------+----------------+---------+
| Brooklyn    | A-201          |     900 |
| Horseneck   | A-215          |     700 |
| Palo Alto   | A-222          |     700 |
+-------------+----------------+---------+
3 rows in set (0.001 sec)
```

No 7 Query
(as Plain Text)

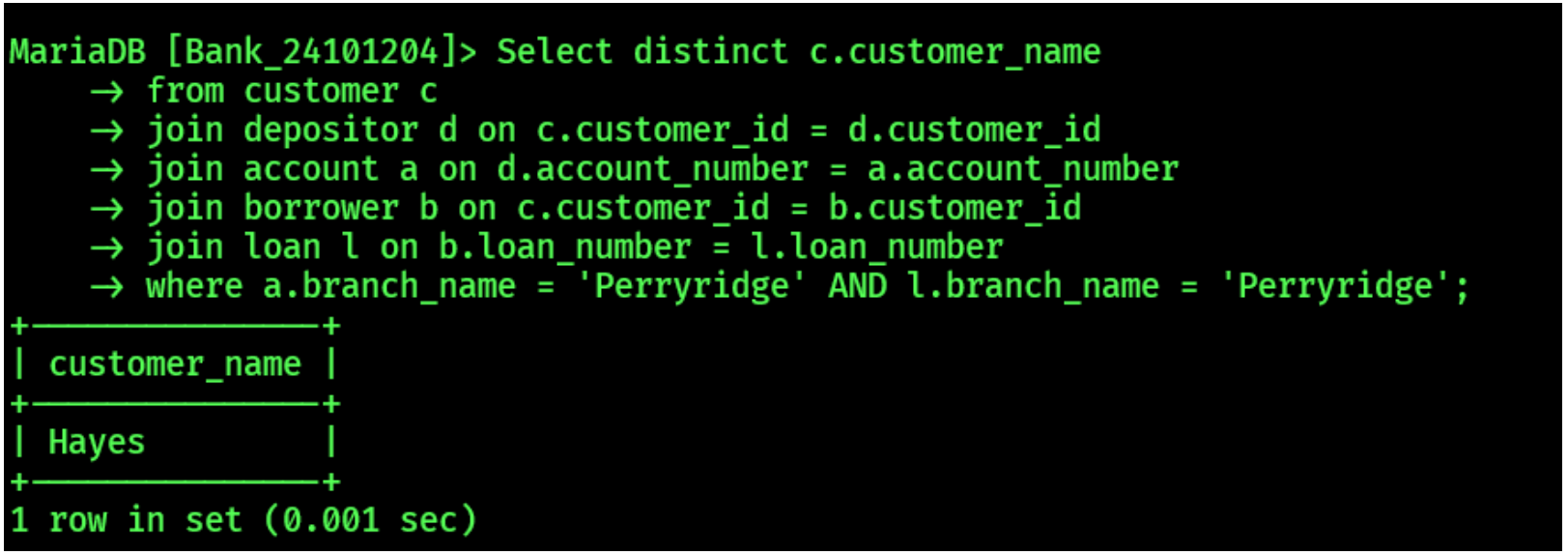
```
select l.loan_number, l.amount, c.customer_name
from loan l
join borrower b ON l.loan_number = b.loan_number
join customer c ON b.customer_id = c.customer_id
order by l.amount desc, l.loan_number desc
Limit 5;
```

No 7 SS
(of Query & Output in Shell)

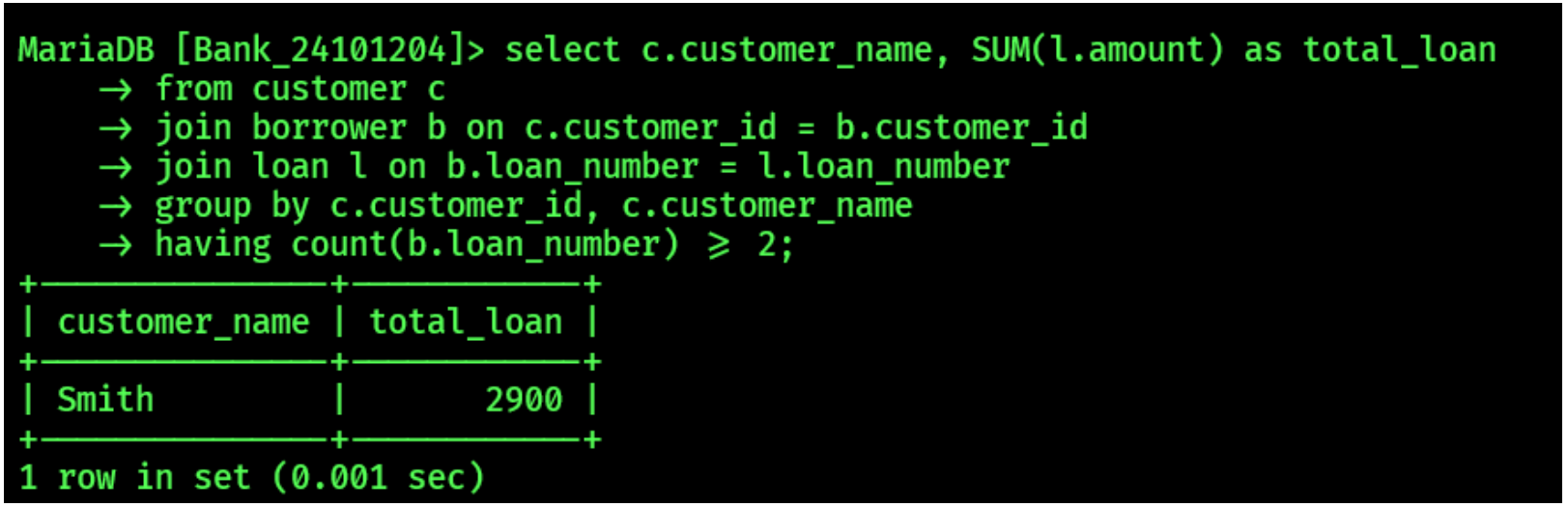
```
MariaDB [Bank_24101204]> select l.loan_number, l.amount, c.customer_name
    -> from loan l
    -> join borrower b ON l.loan_number = b.loan_number
    -> join customer c ON b.customer_id = c.customer_id
    -> order by l.amount desc, l.loan_number desc
    -> Limit 5;
+-------------+--------+---------------+
| loan_number | amount | customer_name |
+-------------+--------+---------------+
| L-23        |   2000 | Smith         |
| L-15        |   1500 | Hayes         |
| L-14        |   1500 | Johnson       |
| L-16        |   1300 | Adams         |
| L-17        |   1000 | Jones         |
+-------------+--------+---------------+
5 rows in set (0.001 sec)
```

No 8 Query
(as Plain Text)

```
Select distinct c.customer_name
from customer c
join depositor d on c.customer_id = d.customer_id
join account a on d.account_number = a.account_number
join borrower b on c.customer_id = b.customer_id
join loan l on b.loan_number = l.loan_number
where a.branch_name = 'Perryridge' AND l.branch_name = 'Perryridge';
```

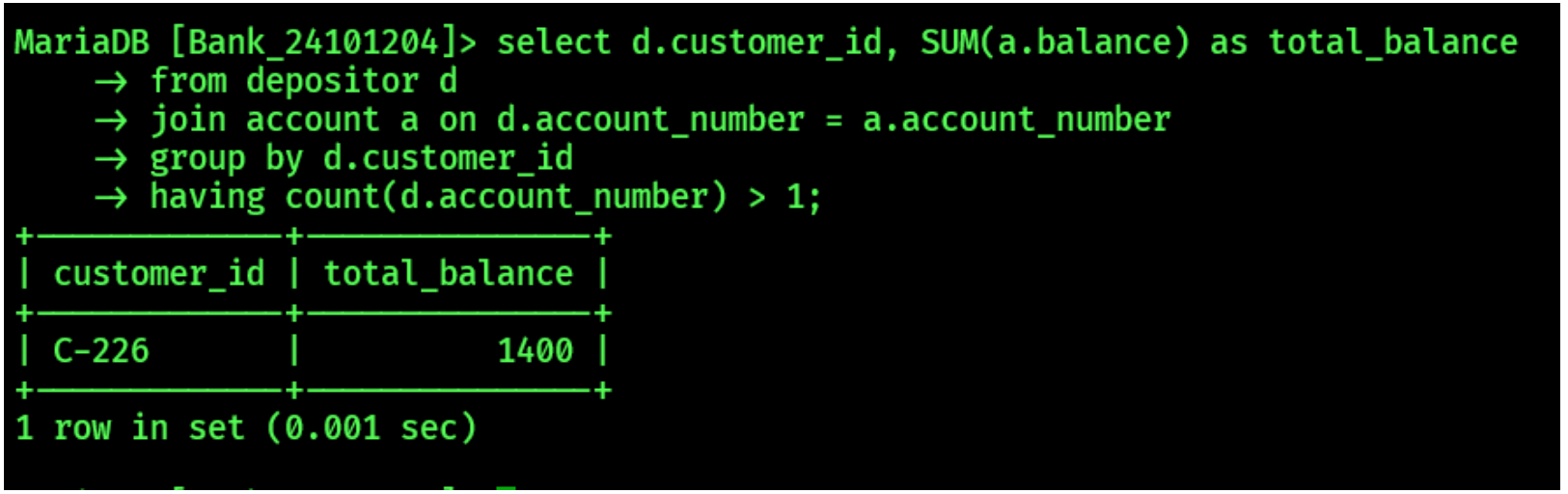
No 8 SS (of Query & Output in Shell)	

No 9 Query (as Plain Text)	`select c.customer_name, SUM(l.amount) as total_loan`
	`from customer c`
	`join borrower b on c.customer_id = b.customer_id`
	`join loan l on b.loan_number = l.loan_number`
	`group by c.customer_id, c.customer_name`
	`having count(b.loan_number) >= 2;`

No 9 SS (of Query & Output in Shell)	
No 10 Query (as Plain Text)	

```
MariaDB [Bank_24101204]> select c.customer_name, SUM(l.amount) as total_loan
    -> from customer c
    -> join borrower b on c.customer_id = b.customer_id
    -> join loan l on b.loan_number = l.loan_number
    -> group by c.customer_id, c.customer_name
    -> having count(b.loan_number) ≥ 2;
+---------------+------------+
| customer_name | total_loan |
+---------------+------------+
| Smith         |       2900 |
+---------------+------------+
1 row in set (0.001 sec)
```

```
select d.customer_id, SUM(a.balance) as total_balance
from depositor d
join account a on d.account_number = a.account_number
group by d.customer_id
having count(d.account_number) > 1;
```

No 10 SS **(of Query & Output in Shell)**	

```
MariaDB [Bank_24101204]> select d.customer_id, SUM(a.balance) as total_balance
    -> from depositor d
    -> join account a on d.account_number = a.account_number
    -> group by d.customer_id
    -> having count(d.account_number) > 1;
+-------------+-----------------+
| customer_id | total_balance |
+-------------+-----------------+
| C-226       |          1400 |
+-------------+-----------------+
1 row in set (0.001 sec)
```